

Helix Thready — Motion Spec & Lottie Set

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Rev	Date	Author	Change
1	2026-07-22	swarm (design/motion)	Initial motion package: duration/easing tokens, six hand-authored Lottie assets, interaction→animation map, reduced-motion contract, runtime integration, honest validation record
2	2026-07-22	swarm (design · motion)	Delivered the five static <code>reducedMotionFallback</code> SVGs (geometry derived from the Lottie keyframes at the contract frames); manifest v2 names them; asset half of <code>THREADY-MOT-03</code> closed, remainder narrowed (\$8); preview.html fallback toggle; validation record extended (\$7.7)

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1. Scope & sources of truth

This package is the concrete motion layer the design area mandates. It does **not** re-decide anything; it implements:

- **design-system.md §5** — the two shipped motion tokens (`--motion-fast 150ms` , `--motion-base 200ms`) on `--ease-standard cubic-bezier(0.2, 0, 0, 1)` , and the rule that **all motion honors prefers-reduced-motion: reduce** [VERIFIED — IN-HOUSE: design_system] .
- **prototypes.md §6** — the extended choreography table (route transitions 250–300 ms, skeleton→content cross-fade, looped processing pulse, toast timing) and the “lots of Lottie animations” requirement with static fallbacks [OPERATOR] .
- **ux-flows.md** — the four journeys each animation binds to (§4).
- Brand color values are the Thready theme tokens (design-system.md §3.2); the error color is the shared `--danger` semantic token — **never** `--accent` — per the a11y contract (design-system.md §6) [VERIFIED] .

Everything under `motion/` is hand-authored vector Lottie (v5.x schema, `ty: 4` shape layers only, zero raster assets). The files partially realize the **Lottie half of** [OPEN: THREADY-DES-02] (Lottie is not a native OpenDesign/Figma export — see §8).

2. Duration & easing tokens

Verified tokens come from `tokens/core.css` ; extended tokens follow prototypes.md §6 and are [DEFAULT — adjustable] .

Token	Value	Provenance	Use
<code>--motion-fast</code>	150 ms	[VERIFIED — <code>core.css</code>]	control state (hover/press), toast exit, theme crossfade
<code>--motion-base</code>	200 ms	[VERIFIED — <code>core.css</code>]	enter/exit, disclosure, drawer/sheet slide, toast enter, skeleton→content
<code>--ease-standard</code>	<code>cubic-bezier(0.2, 0, 0, 1)</code>	[VERIFIED — <code>core.css</code>]	the only easing curve; every eased Lottie keyframe uses it
<code>--motion-route</code>	300 ms	[DEFAULT] (250–300 ms band, prototypes \$6)	route/page shared-axis transition; <code>transition-fade-slide.json</code> runs each card move at exactly 18 f @ 60 fps = 300 ms
<code>--motion-confirm</code>	1 500 ms	[DEFAULT]	one-shot confirmation choreography (<code>success-check</code> , <code>error-cross</code> : 90 f @ 60 fps)
<code>--motion-loop-period</code>	2 000 ms	[DEFAULT]	loop period of the ambient loops (<code>helix-spinner</code> , <code>processing-pulse</code> , <code>thread-sync</code> : 120 f @ 60 fps)

Easing in the Lottie files. Every eased keyframe carries the token curve encoded as Lottie handles — outgoing `{ "x": [0.2], "y": [0] }`, incoming `{ "x": [0], "y": [1] }` — so the assets and the CSS transitions share one acceleration profile. Constant-rate movement (ripple expansion, dot travel across the

thread, the spinner's phase advance between dense samples) is deliberately linear: a ripple or carrier dot that eases reads as stalling.

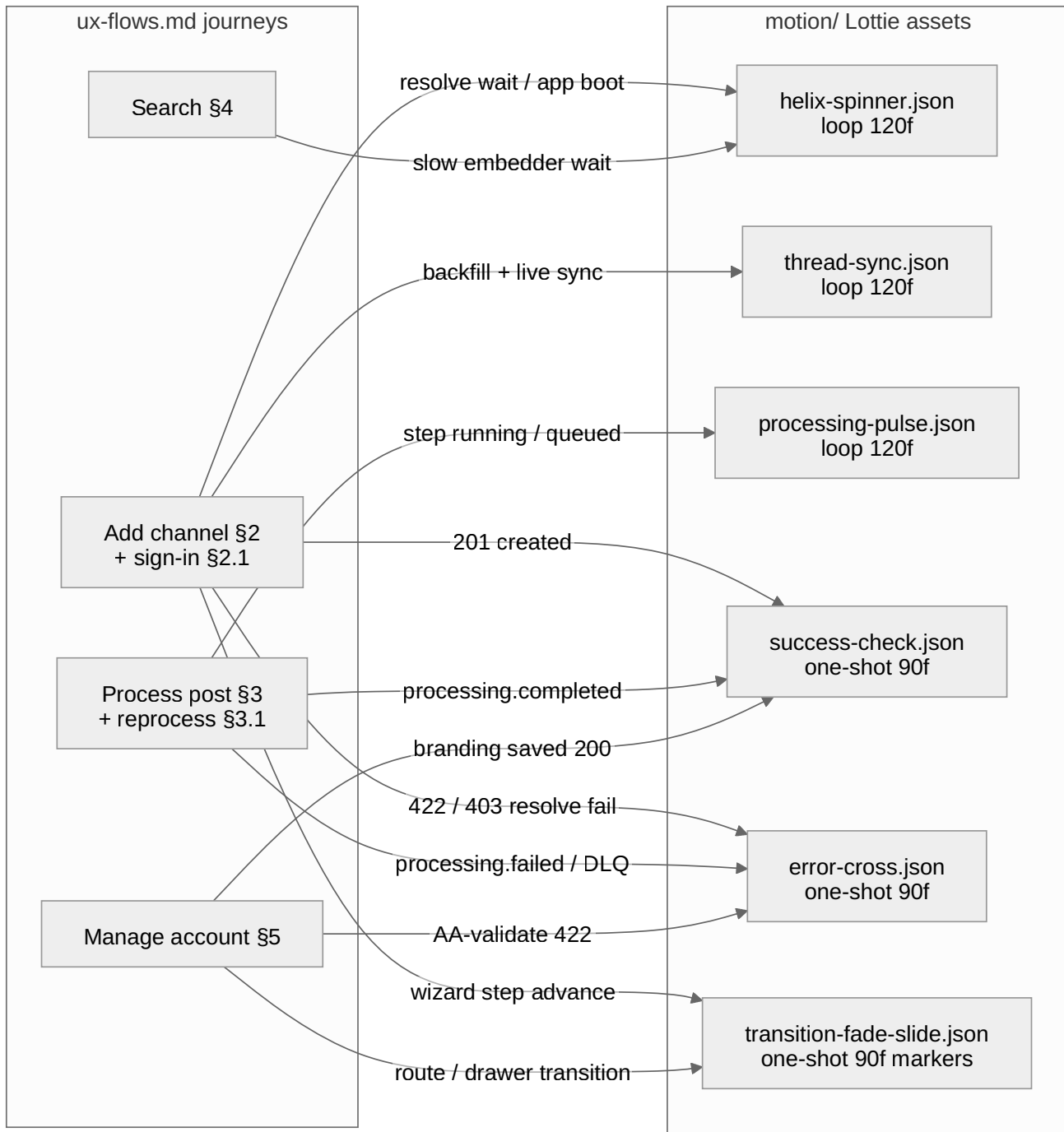
3. The Lottie asset set

All files: schema `v: "5.7.4"`, `fr: 60` declared, `ip: 0`, shape layers only, `assets: []` (no rasters). Colors are the Thready token values in Lottie's 0–1 RGBA form: lime `#B6E376` = `[0.714, 0.890, 0.463]`, accent `#446E12` = `[0.267, 0.431, 0.071]`, teal `#ABDDC9` = `[0.671, 0.867, 0.788]`, teal-dark `#B7EBD6` = `[0.718, 0.922, 0.839]`, brand-ink `#0A0F04` = `[0.039, 0.059, 0.016]`, ink `#020817` = `[0.008, 0.031, 0.090]`, snow `#F8FAFC` = `[0.973, 0.980, 0.988]`, danger `#DC2626` = `[0.863, 0.149, 0.149]`.

File	Canvas	Frames	Loop	Structure
<code>helix-spinner.json</code>	512×512	120 (2 s)	loops , seam-exact	Two sine strands (the double helix seen side-on) as animated bezier paths — 21 phase-sampled shape keyframes per strand advance the phase a full 2π per cycle, which is the helix's rotation; strand opacities counter-pulse (100↔70) so front/back dominance alternates as they weave. Lime strand + teal strand, 30 px round-cap strokes.
<code>success-check.json</code>	512×512	90 (1.5 s)	one-shot	Lime disc pops in (scale 0→108→100, token ease), check draws via trim-path (f22–48), accent ring pings outward (f18–56). Check layer is parented to the disc. Check stroke is <code>--brand-ink #0A0F04</code> on the lime fill — 13.15:1, the documented brand-ink pairing [VERIFIED — design-system §3.2] .
<code>error-cross.json</code>	512×512	90 (1.5 s)	one-shot	Danger disc pops in, two snow cross arms draw sequentially via trim-paths (f24–38, f36–52), disc shakes on X (f54–78), danger ring pings. Arms parented to the disc so pop and shake carry them. Error uses <code>--danger</code> , never brand [design-system §6] .
<code>processing-pulse.json</code>	512×512	120 (2 s)	loops , seam-exact	Lime core breathes (scale 100→114, twice per cycle, token ease); two teal ripple rings expand 100→260 % and fade, staggered 60 f apart. Ring B is ring A phase-shifted with hold-jump keyframes so the loop boundary matches exactly.
<code>thread-sync.json</code>	512×256	120 (2 s)	loops , seam-exact	Messenger node (teal, left) and Thready node (lime, right) joined by a 45 %-opacity teal thread; a lime dot travels → (f0–55) and a teal-dark dot travels ←

File	Canvas	Frames	Loop	Structure
				(f60–115), each fading in/out at the ends; the receiving node bumps 112 % on arrival.
<code>transition-fade-slide.json</code>	512×320	90 (1.5 s)	one-shot, markers	Shared-axis X: outgoing teal card slides –80 px and fades (f15–33), incoming lime card slides in from +80 px (f24–42). Each move is 18 f @ 60 fps = 300 ms — real product timing, not a slowed demo. Lottie markers <code>transition-start</code> (f15, dur 27) and <code>transition-end</code> (f42) delimit the playable segment.

4. Interaction → animation map



Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: `diagrams/motion-asset-map.mmd`.

Explanation (for readers/models that cannot see the diagram). The diagram has two groups. On the left are the four key journeys from `ux-flows.md`: Add channel (with its messenger sign-in sub-flow), Process post (with reprocess), Search, and Manage account. On the right are the six Lottie assets in this directory, each annotated with its loop behavior and frame count. Twelve edges connect journeys to assets. Add channel

uses four: the helix spinner while a link resolve or app boot is pending, the thread-sync loop while the new channel backfills and streams live posts, the success check when the API returns `201 channel`, and the error cross when a resolve fails with `422` / `403`; its wizard step advance uses the fade-slide transition. Process post drives three: the processing pulse for a queued or running pipeline step (including a reprocess in flight), the success check on `processing.completed`, and the error cross on `processing.failed` or a dead-letter surfaced in the UI. Search uses the spinner only for the degraded slow-embedder wait (its normal loading state is a skeleton, not a spinner). Manage account uses the success check on a `200` branding save, the error cross on a `422` AA-validation failure, and the fade-slide transition for route/drawer navigation.

The full interaction table, including the non-Lottie motion the tokens already cover:

Interaction (journey ref)	Animation	Timing	Notes
App boot / route-level load	<code>helix-spinner</code> loop	2 s loop	the “launcher-spiral loader” of prototypes §6; centered, ≤ 128 px
Add channel: <code>channels:resolve</code> pending (ux §2)	<code>helix-spinner</code> small (48 px)	loop	inline next to the Resolve button
Add channel: <code>201</code> created (ux §2)	<code>success-check</code>	1.5 s one-shot	in the wizard’s final step, then auto-advance to the channel row
Add channel: <code>422</code> / <code>403</code> resolve fail	<code>error-cross</code> at 48 px	one-shot, plays once	beside the inline <code>--danger</code> field message; never replaces the text reason
Channel backfill / live sync (ux §2 “sync begins (live)”)	<code>thread-sync</code> loop	2 s loop	in the channel row / sync banner while <code>post.received</code> events stream
Messenger session established (ux §2.1)	<code>success-check</code>	one-shot	on <code>SessionReady → Persisted</code>
Sign-in bad code / bad password (ux §2.1)	none — field-level <code>--danger</code> text only	<code>--motion-fast</code> color	a retry loop is not an event; reserving <code>error-cross</code> for terminal failures

Interaction (journey ref)	Animation	Timing	Notes
			keeps it meaningful
Pipeline step queued/running (ux §3)	<code>processing-pulse</code> loop	2 s loop	per-step in Post detail §3.6 + Dashboard queue rows
Reprocess in flight (ux §3.1)	<code>processing-pulse</code> loop	loop	<code>409</code> already processing shows a toast (slide, <code>--motion-base</code>) — no error-cross; it is a benign race
<code>processing.completed</code> (ux §3)	<code>success-check</code> in the completion toast	one-shot	assets appear with skeleton→content cross-fade (200 ms)
<code>processing.failed</code> / dead-letter (ux §3)	<code>error-cross</code>	one-shot	next to the surfaced retry-step affordance
Search wait (ux §4)	skeleton (CSS), not a spinner	200 ms cross-fade	<code>helix-spinner</code> only appears in the degraded slow-embedder timeout state
Branding save <code>200</code> (ux §5)	<code>success-check</code>	one-shot	with the audit-logged toast
Branding save <code>422</code> AA-fail (ux §5)	<code>error-cross</code> at 48 px	one-shot	beside the contrast-ratio message + suggestion

Interaction (journey ref)	Animation	Timing	Notes
Route/page navigation, wizard steps	<code>transition-fade-slide</code> pattern	300 ms	implement natively (CSS/Angular animations) per surface; the Lottie file is the canonical reference rendering of the curve/choreography and doubles as an onboarding/demo asset
Button/control state, disclosure, toasts, theme toggle	CSS tokens only	150/200 ms	no Lottie — design-system §5

Motion-with-meaning rules (from `ux-flows §6`): one ambient loop per view maximum; `error-cross` is reserved for terminal failures (retries and benign `409` races get text/toast only); confirmations play once and are never required to understand state — the text/status always carries the meaning.

5. Reduced-motion contract (`prefers-reduced-motion`)

Non-negotiable (design-system §5 `[VERIFIED]`, prototypes §6): motion is enhancement, never required for meaning. Under `prefers-reduced-motion: reduce`:

Animation	Reduced-motion behavior	Poster frame
<code>helix-spinner</code>	do not play; render frozen at the poster frame (a static double-helix mark); the wait state is conveyed by text (“Loading...”)	15
<code>success-check</code>	jump straight to the final frame (check fully drawn) — state is shown, nothing moves; an optional 150 ms opacity-only fade is permitted	89
<code>error-cross</code>	jump to the final frame (cross drawn, no shake)	89
<code>processing-pulse</code>	freeze at poster (core mid-breath at 114 %, ring A distinct + ring B faint — the true frame-30 render); progress conveyed by the step label / percentage text	30
<code>thread-sync</code>	freeze at poster; sync progress conveyed by the counter text (“n posts synced”)	30
<code>transition-fade-slide</code>	route/step changes become an instant cut (or ≤ 100 ms opacity-only fade); no translation	89

Reference implementation (web, the same gate `preview.html` uses):

```
const reduced = window.matchMedia('(prefers-reduced-motion: reduce)').matches;
const anim = lottie.loadAnimation({
  container, renderer: 'svg',
  loop: meta.loop, autoplay: !reduced,
  animationData,
});
if (reduced) anim.goToAndStop(meta.posterFrame, /* isFrame */ true);
```

Static fallback SVGs (delivered 2026-07-22). Each playable animation now ships a dedicated `reducedMotionFallback` SVG, named per animation in `motion-manifest.json` (v2) — this closes the asset half of `THREADY-MOT-03` (§8 keeps the narrowed remainder):

Animation	Fallback file	Renders
<code>helix-spinner</code>	<code>spiral-static.svg</code>	the frame-15 poster: strand paths are the exact 50/50 interpolation of the $t = 12/18$ shape keyframes (their easing is linear); strand opacities 73.7 %/96.3 % are the token ease evaluated at the same frame
<code>success-check</code>	<code>check.svg</code>	final frame 89: disc at rest, check fully drawn; the accent ring ping has opacity 0 there and is omitted
<code>error-cross</code>	<code>cross-static.svg</code>	final frame 89: disc back at 256,256 after the shake, both arms fully drawn; invisible ring ping omitted
<code>processing-pulse</code>	<code>pulse-static.svg</code>	the frame-30 poster as actually rendered : core at its keyframed 114 %, ring A at 140 %/45.8 % opacity and ring B faint at 220 %/15.5 % (the “one faint ring” wording above is prose shorthand; the file matches the renderer)
<code>thread-sync</code>	<code>sync-static.svg</code>	the frame-30 poster: nodes + thread at rest, carrier dot at x 365.78 — the token ease <code>cubic-bezier(0.2, 0, 0, 1)</code> evaluated at frame 30 of its 0–55 f travel
<code>transition-fade-slide</code>	<code>none (null)</code>	reduced motion is an <code>instant-cut</code> — no fallback artwork exists by contract

Geometry is **not redrawn** — every path/radius/opacity is computed from the Lottie’s own keyframe data at the contract frame (derivations in each SVG’s header comment). Coloring `[DEFAULT — adjustable]`: `style="fill: var(--brand, #B6E376)"` -form — literal fallbacks are the Lottie’s baked colors so standalone loads match the animations; token-bearing pages theme the inlined SVG via the CSS variables (the SVG analogue of the \$6 load-time remap). `currentColor` was rejected: multi-color assets.

Poster-frame freezing remains the **equivalent runtime alternative** where an image swap is inconvenient (Compose: `LottieAnimation(progress = posterProgress)`; iOS: `currentFrame = poster`) — both paths render the same state by construction.

6. Runtime integration

Runtimes per `prototypes.md §6`: `lottie-web` (web + Tauri 2 desktop wrapping the Angular UI), `lottie-compose` (KMP/Android), `lottie-ios`. The TUI does **not** consume Lottie — its equivalents are Lipgloss spinner/progress primitives themed by the token bridge (`design-system §7`).

- **Loop config:** `helix-spinner`, `processing-pulse`, `thread-sync` → `loop: true` (all three are seam-exact: the frame-120 state equals frame 0, verified in §7). `success-check`, `error-cross` → `loop: false`, play once per event.
- **Segment playback:** `transition-fade-slide` carries markers; play `[15, 42]` (`transition-start` → `transition-end`) when using it as a real overlay, or full range as a demo/onboarding asset.
- **Sizing:** vector — render at any size; intended in-product sizes: spinner 48–128 px, pulse 24–48 px (inline step indicator), check/cross 48–96 px, thread-sync banner height 48–64 px.
- **Theming (honest limitation).** The JSONs carry literal Thready colors — Lottie has no CSS-variable binding. The lime/teal set is tuned for **dark surfaces** (lime is the dark accent, 13.56:1 on `#020817` [`VERIFIED — design-system §3.4`]); on light surfaces lime is decorative (1.47:1) and load-bearing loading states should either sit on `--surface-warm` / a dark stage, or remap lime → `#446E12` at load time. Because every color is a plain `[r, g, b, 1]` array, a load-time remap is trivial and lossless:

```
// Light-theme remap: lime -> AA accent, before loadAnimation. [DEFAULT — adjustable]
const LIME = [0.714, 0.890, 0.463], ACCENT_LIGHT = [0.267, 0.431, 0.071];
function retint(node: unknown): void {
  if (Array.isArray(node)) { node.forEach(retint); return; }
  if (node && typeof node === 'object') {
    const p = node as Record<string, unknown>;
    const k = (p['c'] as Record<string, unknown> | undefined)?.['k'];
    if ((p['ty'] === 'fl' || p['ty'] === 'st') && Array.isArray(k)
      && LIME.every((v, i) => Math.abs((k[i] as number) - v) < 1e-3)) {
      (k as number[]).splice(0, 3, ...ACCENT_LIGHT);
    }
    Object.values(p).forEach(retint);
  }
}
```

- **White-label:** the same remap is the per-Account brand hook (`theming`) — replace lime/teal with the Account's `--brand` / `--brand-2` at load; the error cross always keeps `--danger` (semantic, never white-labeled).

7. Validation record (honest)

Executed on 2026-07-22 on the authoring host (scripts in the session scratchpad; deterministic generator, re-runnable):

1. **JSON parse** — `python3 json.load` over all six files: **PASS** (all parse).
2. **Structural lint** (custom validator): required top-level keys (`v fr ip op w h layers`) present; `v` is 5.x; `fr` = 60 declared; frame counts 90–120 (within the 60–180 requirement); every layer is `ty: 4` with a non-empty `shapes` array and a full `ks` transform; keyframe times strictly monotonic; every color component within [0, 1]; `assets` empty (no rasters); and for the three loopers a **loop-seam check** — the rendered state at `ip` equals the state at `op` (properties may differ only where the owning layer has opacity 0 at both edges): **PASS, all six**. Two genuine defects were caught and fixed by this gate during authoring (a 1-ulp floating-point seam mismatch in the spinner's phase wrap; an over-strict seam rule superseded by the visibility-aware one).
3. **Real renderer** — **lottie-web 5.13.0** (SVG renderer) headless under jsdom (with a canvas-context stub; the SVG renderer does not draw to canvas): every file loads, seeks frames 0 / ¼ / ½ / end without exceptions and produces real SVG path geometry: **PASS, all six**.
4. **Animation-progression assertions** under the same renderer: the spinner strand path **morphs** between frames; the check and cross trim-paths **grow** while drawing; the pulse ring transform **expands** (scale 1.13 → 2.33 between f10 and f100); the fade-slide card **translates and fades** (x 277 → 256, opacity 0.73 → 1 between f30 and f42); thread-sync node bump **scales** on arrival: **PASS**.
5. **preview.html smoke test** — the assembled page booted under jsdom (matchMedia polyfilled — a jsdom gap, present in all real browsers): 6 cards, 6 live SVG stages, 27 rendered paths: **PASS**.
6. **Real-browser render (headless Chromium, Blink/Skia compositor)** — a contact sheet of all six files at six frame positions on light **and** dark stages, plus the assembled `preview.html` in light, explicit-dark (`data-theme="dark"`), and forced `prefers-reduced-motion` modes, was screenshot-rendered and reviewed: the strands weave and swap depth, the check/cross trim-paths draw, rings expand, sync dots shuttle both directions, cards cross-fade on axis; under reduced motion every asset loads frozen on its poster frame with the explanatory note shown; no console errors: **PASS**.
7. **Static fallback SVGs (rev 2, 2026-07-22)** — the five `reducedMotionFallback` files: values (interpolated strand paths, eased opacities, dot position, ring scales) computed from the Lottie keyframe data with the file's own easing (token curve / linear as encoded), **not** eyeballed; all five pass `xmllint --noout`; `preview.html` re-smoke-tested under jsdom after the fallback-toggle addition in both normal and forced-reduced-motion modes: 6 cards, 6 live SVG stages, 27 rendered paths (unchanged from item 5), 5 inline fallback SVGs + the instant-cut note card, fallback view auto-engaged under reduced motion, toggle flips both ways, no errors: **PASS**.

Not done (stated plainly): no review on a **physical display** (headless Chromium screenshots only); no verification in `lottie-compose` / `lottie-ios` / skottie (renderer-conformance differences are possible, though only widely-supported features are used: shape layers, trim paths, hold keyframes, parenting,

markers); no visual-regression baseline captured yet (`ScreenDiff` bank, `[GAP: 9.3]`); the five fallback SVGs have **not** had an independent design review or a pixel-level A/B against the frozen Lottie frames (they match by construction, not by screenshot diff) — kept open in `THREADY-MOT-03` (§8); the rev-1 Chromium screenshot pass predates the preview’s fallback toggle.

8. Gaps & open items

- `[OPEN: THREADY-DES-02]` (Lottie half) — **partially delivered**: this set proves hand-authored token-true Lottie without the AE/ `bodymovin` path and covers the functional states. Still open for **illustration-grade** assets (empty-state illustrations, onboarding accents per prototypes §6), which need the design-tool bridge. The `spiral-loader` / `processing-pulse` / `success-check` ids of the prototypes §6 manifest map to `helix-spinner` / `processing-pulse` / `success-check` here (see `motion-manifest.json`).
 - `[OPEN: THREADY-MOT-01]` — verify all six files in `lottie-compose` and `lottie-ios` before mobile ships them.
 - `[OPEN: THREADY-MOT-02]` — decide light-theme strategy: load-time remap (§6) vs. shipping generated `-light` variants; wire the white-label remap into the branding editor preview.
 - `[OPEN: THREADY-MOT-03 — narrowed 2026-07-22]` — the **asset half is closed**: the five static `reducedMotionFallback` SVGs are delivered, named in the manifest (v2), keyframe-derived and xmllint-validated (§5, §7.7). Still open, narrowed to:
 1. independent design review / pixel A-B of the five statics against the frozen Lottie frames on a physical display; (b) the `empty-channels.svg` fallback from the prototypes §6 excerpt — blocked on the illustration-grade assets of `[OPEN: THREADY-DES-02]` , delivered together with them.
 - `[GAP: 9.3]` — add these animations (poster frames per theme) to the `ScreenDiff` visual-regression bank once its CI exists.
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